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LSTM Comparative Study

Machine Learning



A Comparative Study of Attention-Enhanced LSTM Architectures for Turbofan Remaining Useful Life Prediction Using NASA C-MAPSS FD004

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1. Introduction

Accurate prediction of Remaining Useful Life (RUL) is important in **predictive maintenance**, especially in **safety-critical industries**. For turbofan engines, knowing how many cycles remain before failure helps operators plan maintenance early, reducing both **unexpected failures** and **unnecessary repairs**. The problem is uneven because **overestimating RUL** can lead to dangerous failures, while **underestimating RUL** can waste time and maintenance costs. Therefore, a good prognostic model should be evaluated not only on **overall accuracy** but also on how well it predicts engines **close to failure**.

Long Short-Term Memory (LSTM) networks are widely used for RUL prediction because they can learn patterns from **time-series sensor data**. However, many different LSTM architectures exist, including **stacked LSTMs**, **bidirectional LSTMs**, and **attention-based models**. These designs are often chosen without detailed comparison, making it unclear which changes actually improve performance.

This report compares **five LSTM-based models** using the **NASA C-MAPSS FD004 dataset**. Each model adds only **one new feature** compared to the previous model, allowing a clear understanding of how each architectural change affects performance. The results from this study are also compared with findings from previous research by Peringal and Kimollo & Liu to identify trends that remain consistent across studies.

2. Dataset and Preprocessing

2.1 NASA C-MAPSS FD004

The **Commercial Modular Aero-Propulsion System Simulation (C-MAPSS)** dataset released by NASA is a widely used benchmark for **turbofan engine RUL prediction**. This study uses the **FD004 subset**, which is the most difficult of the four datasets. It contains **249 training engines** and **248 test engines** operating under **six different conditions** and includes **two fault types: High-Pressure Compressor (HPC) degradation** and **fan degradation**. This makes FD004 more challenging than **FD001**, which has only one operating condition and one fault mode. Therefore, RMSE values from FD001 and FD004 should not be directly compared.

Each engine contains a **run-to-failure sequence** with **21 sensor measurements** and **3 operational settings** recorded per cycle. The features were reduced from **24 to 13** by removing sensors with almost no variation, since they provide little useful information. A **5-cycle rolling mean** was applied to reduce noise while keeping the degradation pattern clear.

To handle different operating conditions, **KMeans clustering (k=6)** was used on the operational settings. After clustering, a separate **StandardScaler** was applied for each condition using training data and then used on both training and test sets. This **per-condition normalisation** is important because the same sensor values can represent different engine states under different operating conditions.

2.2 Sequence Construction and RUL Labels

Sliding windows of length 30 cycles (stride=1) were constructed per engine, yielding 54,028 training windows and 34,081 test windows, each of shape (30, 13). RUL labels for training were computed as $\text{max_cycle} - \text{current_cycle}$, clipped at 125 cycles to produce a piecewise-linear target that avoids the model being distracted by the early healthy phase where all engines look similar.

A critical property of this labelling scheme is the resulting class imbalance: **42.7% of training windows sit at the plateau value of RUL = 125**, while only **14.3% are in the operationally critical**

danger zone ($RUL \leq 30$ cycles). This imbalance systematically biases all models toward optimistic predictions and is the primary driver of the positive prediction bias observed across all five architectures in this study.

3. Model Architectures

Five architectures are evaluated in a progressive augmentation design. All share the same fixed hyperparameters:

LSTM hidden units = 64, dropout rate = 0.2, batch size = 256, Adam optimiser with initial learning rate 10^{-3} , EarlyStopping (patience = 15, restoring best weights), and ReduceLROnPlateau (patience = 6, factor = 0.5). Keeping these constant ensures that performance differences are attributable to architectural changes rather than hyperparameter choices.

ID	Name	Architecture	Params	Loss
M1	Vanilla Basic LSTM	Input $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dense(1)	~22K	MSE
M2	Stacked LSTM	Input $\rightarrow$ LSTM(64, seq) $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32) $\rightarrow$ Dense(1)	~55K	MSE
M3	Stacked BiLSTM	Input $\rightarrow$ BiLSTM(64, seq) $\rightarrow$ BiLSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32) $\rightarrow$ Dense(1)	~143K	MSE
M4	BiLSTM + Attn	Input $\rightarrow$ BiLSTM(64, seq) $\rightarrow$ BiLSTM(64, seq) $\rightarrow$ Bahdanau Attn $\rightarrow$ Dense(64) $\rightarrow$ Dense(32) $\rightarrow$ Dense(1)	~157K	MSE
M5	BiLSTM + Attn + W.Loss	Same as M4 (architecture unchanged)	~157K	Weighted MSE

M1 — Vanilla LSTM (Baseline)

The single-layer LSTM reads the 30-cycle sensor window sequentially. At the final timestep, the hidden state h_{30} is passed to a Dense(32) $\rightarrow$ Dense(1) regression head. With only ~22K parameters, this model is deliberately minimal. It defines the performance floor: any subsequent architecture that fails to outperform it is not justified by its added complexity.

M2 — Stacked LSTM

A second LSTM layer is added, with the first configured to return its full hidden-state sequence (return_sequences=True). This enables hierarchical temporal processing: the first layer captures cycle-level dynamics, and the second processes the resulting sequence to learn longer-range degradation patterns. Parameter count increases to ~55K. The hypothesis is that depth improves capture of multi-step degradation dynamics, but greater capacity also increases the risk of overfitting the plateau-heavy training distribution.

M3 — Stacked Bidirectional LSTM

Bidirectional wrapping runs two LSTM passes — forward (cycles 1 $\rightarrow$ 30) and backward (cycles 30 $\rightarrow$ 1) — concatenating their outputs at each timestep. This doubles the per-timestep representation dimension to 128 and allows each timestep to incorporate context from both ends of the window. Because the full 30-cycle window is always available at inference time, using backward context is not

a look-ahead violation. The backward pass is particularly valuable in the FD004 multi-condition setting, where early-window readings are better interpreted in light of how the sensor values evolve by the end of the window.

M4 — Stacked BiLSTM with Bahdanau Attention

Both BiLSTM layers are configured to return full sequences, and a Bahdanau (additive) attention layer is added. Rather than using only the final BiLSTM output, attention computes a softmax-normalised score over all 30 timestep representations to form a context vector $c = \sum \alpha_i h_i$. This context vector replaces the final hidden state as input to an expanded regression head (Dense(64) → Dense(32) → Dense(1)). The attention weights α_i are fully interpretable and can be visualised per prediction, showing which cycles the model emphasised.

M5 — Stacked BiLSTM with Attention and Weighted Loss

Architecturally identical to M4. The sole change is the training objective: MSE is replaced by failure-region weighted MSE, where the squared error for each training window is multiplied by a weight $w = 1 + (30 - RUL)/30$ when $RUL \leq 30$ (rising linearly from 1.0 at $RUL=30$ to 2.0 at $RUL=0$), and $w = 1.0$ otherwise. This directly amplifies the gradient signal for end-of-life windows, aligning the training objective with the asymmetric NASA evaluation metric.

4. Evaluation Methodology

Four metrics are computed for each model, all using the last observed window per engine as the prediction point — consistent with the NASA PHM 2008 challenge specification.

- **RMSE (Root Mean Squared Error):** the primary accuracy metric, computed in RUL cycles. Penalises large errors quadratically.
- **MAE (Mean Absolute Error):** the average absolute error in cycles; less sensitive to outliers than RMSE.
- **NASA Score (Standard):** the official asymmetric scoring function. For each engine, $d = \text{predicted} - \text{true RUL}$. The penalty is $\exp(d/10) - 1$ for $d \geq 0$ (late, dangerous) and $\exp(-d/13) - 1$ for $d < 0$ (early, conservative). The 10/13 asymmetry means a 20-cycle late prediction incurs approximately 2.4× the penalty of a 20-cycle early prediction. Summed across all engines; lower is better.
- **NASA Score (Modified):** a variant developed for this study. The standard per-engine penalty is amplified by $w = 1 + (30 - RUL)/30$ for engines with true final $RUL \leq 30$ cycles. The Mod/Std ratio serves as a diagnostic: values above 1.10 indicate errors are disproportionately concentrated in the danger zone.

Engine boundary detection: in this study, engine boundaries were reconstructed from the raw test file (test_FD004.txt), enabling evaluation over all 237 engines with test sequences long enough to produce at least one sliding window. A fallback heuristic (RUL-jump detection) recovers approximately 176 engines when the raw file is unavailable, and is noted as a limitation when used.

5. Experimental Results

5.1 Summary Table

Model	RMSE	MAE	NASA	NASA-Mod	Bias	Epochs	Params
M1 — Vanilla LSTM	22.051	17.145	3,204	3,733	+6.10	34	22K

Model	RMSE	MAE	NASA	NASA-Mod	Bias	Epochs	Params
M2 — Stacked LSTM	24.720	19.239	5,727	6,108	+11.72	29	55K
M3 — Stacked BiLSTM	24.587	19.110	5,514	5,714	+12.23	25	143K
M4 — BiLSTM + Attention	23.053	17.438	3,363	3,573	+5.44	23	157K
M5 — BiLSTM + Attn + W.Loss	22.342	17.472	2,735	2,917	+4.36	24	157K

Table 1: Metrics computed on the last observed window per engine. RMSE and MAE in RUL cycles. Bias = mean signed error (predicted – true); positive indicates systematic overestimation. All lower values are better.

5.2 Step-by-Step Analysis

M1 → M2: Stacking Degrades Performance

Adding a second LSTM layer **increased RMSE by 12.1%** (22.051 → 24.720), more than doubled NASA-Std (3,204 → 5,727), and caused prediction bias to nearly double (+6.10 → +11.72 cycles). This is a clear regression, not a marginal one.

The mechanism is overfitting in the presence of plateau dominance. M2's parameter count (55K) is 2.5 times M1's (22K), but 42.7% of the training signal is locked at RUL=125. The second LSTM layer learns a strong prior toward predicting high RUL values — exploiting the dominant class — without gaining meaningful discriminative power for degradation detection. EarlyStopping restored the best validation weights at epoch 29, but even the best point during training represented a worse generalisation than M1's simpler solution. Depth alone, without a mechanism to direct capacity toward informative examples, is counterproductive.

M2 → M3: Bidirectionality Provides Marginal Recovery

Replacing stacked LSTM with stacked BiLSTM improved RMSE slightly (24.720 → 24.587) and NASA-Std from 5,727 to 5,514 — a modest improvement, but both models remain substantially worse than M1. Bias worsened marginally to +12.23 cycles.

Bidirectionality enriches per-timestep representations, which is theoretically beneficial for FD004's multi-condition structure. However, the training dynamics remain dominated by plateau windows. With 143K parameters — the most of any model except M4/M5 — M3 has more capacity than M2 to exploit the plateau-prediction shortcut, and only the richer per-timestep representations slightly counteract this. The conclusion is that bidirectionality is a genuine improvement, but its effect is masked by plateau dominance when the aggregation mechanism is still a single final hidden state.

M3 → M4: Attention Achieves Major Recovery

The addition of Bahdanau attention is the **single largest improvement in the study**. RMSE dropped from 24.587 to 23.053, NASA-Std fell 39% from 5,514 to 3,363, and bias reduced dramatically from +12.23 to +5.44 cycles. M4 achieves the best RMSE of any model in the study.

The mechanism is a fundamental change in sequence aggregation. M2 and M3 compress the 30-cycle window into a single fixed-size vector at the final timestep — all information from earlier cycles must be retained through 30 steps of LSTM gating. Attention instead forms a weighted average over all 30 timestep representations, with weights α_i learned to concentrate on the most diagnostically informative cycles. In practice, attention weights concentrate on the most recent cycles in the window, where degradation signals are strongest and most distinct from the plateau-like early

readings that drive bias. The bias reduction from +12.23 to +5.44 is precisely the consequence: by down-weighting early-window plateau-like readings, the model is less pulled toward predicting high RUL values.

M4 → M5: Weighted Loss Achieves Best NASA Score

With identical architecture, replacing MSE with failure-region weighted MSE reduced NASA-Std by 18.7% (3,363 → 2,735) and bias from +5.44 to +4.36 cycles. RMSE changed marginally (23.053 → 22.342). M5 achieves the best NASA-Std and NASA-Mod of all five models.

The mechanics are as expected: by applying up to 2× gradient weight for end-of-life windows ($RUL \leq 30$), the model allocates more representational capacity to the danger zone during training. The small RMSE improvement reflects that RMSE treats all engines equally regardless of their final RUL, while the NASA score specifically rewards accuracy near failure. M5 trades marginal mid-life accuracy for demonstrably better end-of-life accuracy — a trade that the NASA scoring function recognises and that a maintenance system would value. The persistence of positive bias (+4.36 cycles even with the weighted loss) confirms that the plateau dominance cannot be fully overcome with a 2× weight alone.

5.3 Key Findings

- Architectural depth without attention is counterproductive on this plateau-heavy dataset. M2 and M3 both regress from M1 before attention rescues them.
- Bahdanau attention is the critical enabling component. Its selective aggregation mechanism converts the stacked BiLSTM's representational capacity into genuine discriminative power.
- Loss function alignment with evaluation metrics produces measurable gains without changing model architecture or inference cost.
- All five models exhibit positive prediction bias (+4.36 to +12.23 cycles), which is a dataset-level artifact from the 42.7% plateau windows — not an architectural failure.
- M5 (BiLSTM + Attention + Weighted Loss) is the recommended architecture: best NASA-Std (2,735), best NASA-Mod (2,917), and lowest bias.

6. Comparison with Published Research

6.1 Overview of Reference Studies

Two published studies on LSTM-based RUL prediction are used as cross-references. It is important to note that both use FD001 — the simplest C-MAPSS subset (100 engines, 1 operating condition, 1 fault mode) — while this study uses FD004. Direct RMSE comparisons are therefore not valid, as FD001 typically yields RMSE values around 13–16 cycles while FD004's harder structure produces values around 22–25 cycles with similar architectures. The comparison is instead about which architectural patterns produce consistent findings across experimental contexts.

6.2 Peringal et al. (2024) — Vanilla LSTM vs MLP

Peringal et al. compare a single-layer LSTM against a Multilayer Perceptron on FD001 using PyTorch, min-max feature scaling, and 20-timestep input windows. Their LSTM achieves a test MSE of 796 against the MLP's 1,745 — a 54% improvement. The central finding aligns directly with this study's

baseline: a single-layer LSTM captures temporal degradation patterns that static models cannot, and this advantage compounds as the degradation approaches failure (where temporal context becomes most informative).

A notable methodological difference is their choice of exponentially weighted moving average for noise smoothing versus the rolling mean used here. Both achieve similar noise reduction, but EWA is sensitive to the smoothing factor and can distort recent readings. Their min-max scaling is also global (across all engines) rather than per-condition, which is sufficient for FD001's single operating condition but would be inadequate for FD004's six. The study validates that vanilla LSTM is a strong baseline, but does not explore any architectural enhancements.

6.3 Kimollo & Liu (2024) — CNN, BiLSTM, GRU, and Regression Models

Kimollo & Liu conduct a broader benchmark on FD001, comparing classical regression models (Random Forest, XGBoost, SVM, etc.), deep recurrent models (LSTM, GRU, RNN, Bidirectional LSTM), and CNN-based models. Their key results are instructive. Among recurrent models, Bidirectional LSTM achieves the best training RMSE (12.29) while GRU edges it on test RMSE (14.78 vs 15.14). CNN-4 (two convolutional layers + two max-pooling layers) achieves the best overall test RMSE of 14.02 with $R^2 = 0.88$.

The finding that BiLSTM outperforms standard LSTM in training RMSE **directly corroborates this study's M3 vs M1 comparison**: bidirectionality provides richer temporal representations. However, Kimollo & Liu find that GRU outperforms BiLSTM on test RMSE (14.78 vs 15.14), suggesting that on FD001's simpler structure, the additional parameters of BiLSTM may not fully pay off. On FD004's harder structure with 6 operating conditions and 2 fault modes, bidirectionality should theoretically matter more — a hypothesis supported by M4's strong performance in this study.

Their CNN-4 outperforming all recurrent models is a significant finding: convolutional architectures can extract local temporal patterns (feature gradients over short windows) that recurrent models sometimes miss, especially without attention. A natural extension of this study would be to benchmark a CNN+BiLSTM hybrid against M4 and M5, following the CNN-1+LSTM architecture they also test (test RMSE 14.63).

6.4 Cross-Study Summary

Study	Dataset	Architecture	Best RMSE	Key Finding
Peringal et al. (2024)	FD001 (100 eng)	Vanilla LSTM vs MLP	MSE 796	LSTM captures temporal patterns; MLP underfits time-series
Kimollo & Liu (2024)	FD001 (100 eng)	CNN-4, BiLSTM, GRU, LSTM	14.02 (CNN)	CNN outperforms recurrent models; BiLSTM best among RNNs
This Study — M1	FD004 (249 eng)	Vanilla LSTM (baseline)	22.051	Competitive single-layer baseline; NASA-Std 3,204

Study	Dataset	Architecture	Best RMSE	Key Finding
This Study — M4	FD004 (249 eng)	BiLSTM + Bahdanau Attn	23.053	Best RMSE among non-weighted models; 39% NASA gain over M3
This Study — M5	FD004 (249 eng)	BiLSTM + Attn + W.Loss	22.342	Best NASA-Std (2,735); lowest bias (+4.36); recommended model

Table 2: Cross-study comparison. Note that FD001 and FD004 are not directly comparable by RMSE; architectural findings (which model types perform best) are the transferable insight.

Three patterns are consistent across all studies: (1) temporal models outperform static models; (2) bidirectionality improves over unidirectional LSTM; (3) the single-layer LSTM baseline is more competitive than its simplicity suggests. The unique contribution of this study is the demonstration that attention and loss function alignment, which neither reference study explores, are the most impactful improvements on a hard multi-condition dataset like FD004.

7. Architectural Recommendations for RUL Prediction

7.1 What Works

Based on the combined evidence from this study and the published literature, the following architectural choices are well-supported for turbofan RUL prediction:

- Bahdanau (additive) attention on top of stacked BiLSTM. This is the single most impactful architectural component in this study, producing a 39% NASA score improvement over an equivalent architecture without attention. Its advantage is twofold: it enables selective focus on the most informative cycles (typically the most recent, where degradation signals are sharpest), and it provides interpretable attention weights that can be audited per prediction. For a prognostic system where explainability matters, this is substantive.
- Bidirectional LSTM layers. Consistent findings across this study (M3 outperforms M2, M4 outperforms hypothetical unidirectional attention) and Kimollo & Liu (BiLSTM outperforms standard LSTM in training) confirm that backward context within a fixed window is beneficial. The gain is modest alone but becomes large in combination with attention.
- Failure-region weighted loss functions. M4 → M5 demonstrates that loss function alignment with evaluation metrics produces significant NASA score gains (18.7%) with zero architectural change. Any model deployed in a safety-critical context should use a loss function that reflects the cost asymmetry of the application.
- Per-condition normalisation. For multi-condition datasets like FD004, normalising per operating regime is essential. Global normalisation conflates regime-specific sensor baselines and produces systematically worse models on harder subsets.
- Single-layer LSTM as a robust baseline. All three studies confirm that vanilla LSTM, properly regularised, is harder to beat than expected. On FD001, Peringal et al. achieve lower MSE than the MLP with a single layer; in this study, M1 outperforms M2 and M3.

7.2 What Does Not Work (or When It Does Not Work)

- Naive stacking without attention. M2 and M3 both regress from M1 despite having 2.5–6.5× more parameters. On plateau-heavy datasets, additional LSTM layers learn to exploit the

dominant class rather than improving degradation discrimination. Stacking is only beneficial when paired with an aggregation mechanism (attention) that prevents the extra capacity from being wasted on plateau-prediction shortcuts.

- Global normalisation on multi-condition data. Not tested explicitly here, but inferred from the preprocessing literature: applying a single scaler across all operating conditions conflates regime differences and should be avoided for FD004 and any multi-regime dataset.
- High oversampling ratios for danger-zone windows. Prior experiments referenced in the notebook with $3\times$ oversampling of $RUL \leq 30$ windows caused convergence instability. The $2\times$ maximum weight used in M5's loss function achieves a balance between emphasis and stability. Aggressive oversampling requires careful learning rate and batch size tuning.
- Standard MSE loss for safety-critical prognostics. MSE treats a 20-cycle error at $RUL=5$ the same as a 20-cycle error at $RUL=100$. For turbofan maintenance, these have completely different operational consequences. Any deployed system should use an asymmetric or zone-weighted loss.

7.3 Architecture Selection Guide

For practitioners selecting an LSTM-based architecture for turbofan RUL prediction, the findings suggest a practical decision hierarchy:

- If interpretability and end-of-life accuracy are both required (production prognostics): use BiLSTM + Bahdanau Attention + weighted loss (M5 equivalent). This is the recommended configuration.
- If RMSE is the primary metric and computational budget is limited: vanilla LSTM (M1 equivalent) is often competitive with architectures $6\times$ larger and trains faster. It should be the first model trained, not skipped.
- If the dataset has a single operating condition (FD001-like): CNN-based architectures or GRU are worth benchmarking alongside BiLSTM+Attention; Kimollo & Liu's results suggest CNN can outperform pure recurrent models on simpler datasets.
- Avoid pure stacked LSTM/BiLSTM without attention when the training data has a dominant class (e.g., plateau at RUL cap). The extra capacity will overfit the dominant class without a selective aggregation mechanism.

8. Limitations

- Cross-subset comparison: the reference studies use FD001 while this study uses FD004. RMSE values are not directly comparable. Architectural conclusions are generalisable, but specific numerical targets are not.
- Single random seed: all five models were trained with seed 42. A rigorous comparison would average results across multiple seeds to characterise variance. Training neural networks involves stochastic elements that can shift metrics by several percent.
- Uniform hyperparameters: all models use 64 LSTM units, 0.2 dropout, and $lr=10^{-3}$. These may not be optimal for all five architectures; larger models (M3–M5) might benefit from lower learning rates or higher dropout. Individually tuned hyperparameters could shift the relative rankings.
- Positive bias persistence: the +4.36 cycle bias in M5 represents systematic overestimation of remaining life even after weighted loss training. This is not solved by any architecture studied and remains an open challenge with the piecewise-linear RUL labelling approach.
- No CNN or Transformer comparison: Kimollo & Liu's CNN-4 achieves better RMSE than any recurrent model on FD001. A Transformer-based architecture (BERT-style with self-attention) was not evaluated here and represents a natural extension.

9. Conclusion

This study conducted a controlled ablation of five LSTM-based architectures for turbofan RUL prediction on NASA C-MAPSS FD004, cross-referenced against two published studies that used simpler dataset subsets. The central findings are:

- Bahdanau attention on a stacked BiLSTM backbone is the most impactful single architectural component, reducing NASA-Std by 39% (M3 → M4) through selective temporal aggregation. Its interpretability benefit — visualisable attention weights per prediction — is an additional operational advantage.
- Naive architectural depth (stacked LSTM, stacked BiLSTM without attention) is counterproductive on plateau-heavy training distributions. M2 and M3 both perform worse than the single-layer baseline M1, confirming that more parameters without a better aggregation strategy exploits the dominant class rather than improving degradation discrimination.
- Loss function design is as important as architecture for safety-critical deployment. Failure-region weighted MSE improves NASA-Std by 18.7% (M4 → M5) with no architectural change, demonstrating that gradient signal allocation during training is a design parameter of comparable importance to model structure.
- Published studies on FD001 (Peringal et al., 2024; Kimollo & Liu, 2024) corroborate the findings that vanilla LSTM is a strong baseline and bidirectionality improves over standard LSTM. Neither study explores attention or weighted losses, which are the most impactful interventions identified here.
- M5 (Stacked BiLSTM + Bahdanau Attention + Failure-Region Weighted MSE) is the recommended architecture for multi-condition turbofan RUL prediction, achieving the best NASA-Std (2,735), best NASA-Mod (2,917), and lowest prediction bias (+4.36 cycles) of all models evaluated.

Future directions include: evaluation of CNN+BiLSTM hybrid architectures motivated by the CNN results of Kimollo & Liu; Transformer-based approaches that could improve long-range temporal modelling beyond 30-cycle windows; asymmetric loss functions that penalise late predictions across the full RUL range rather than only in the danger zone; and per-model hyperparameter tuning to isolate architecture effects from capacity effects.

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